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Lab 4.3 MTConnect Adapter Simulation

3.1 MTConnect Adapter

We have not discussed details of MTConnect adapter yet. However, to see the data stream flow of MTConnect agent, we will simulate MTConnect adapter. The simulation data (Sensor_log.txt) and simulation adapter (run_scenario.rb) programmed by ruby can be downloaded from the Github repo here. Please download two files on Raspberry Pi. Please note that two files must be in the same directory in this example. The schematic of the simulation is shown in Figure 8. Raspberry Pi will run MTConnect adapter simulator and data streams are transmitted to the MTConnect agent of laptop. The data log file ('Sensor_log.txt') using Lab1 and Lab2 codes was created by collecting both ADXL345 sensor data (3-axis accelerations), DS18B20 sensor(temperature), and Virtual sensor (humidity). Three sensors have different sampling frequencies. In this scenario, sampling periods are 1 second, 3 seconds, and 3 seconds, respectively. The length of simulation data is 3 minutes, but we will simulate repeating the data continuously. The plots of the data are shown in Figure 9.

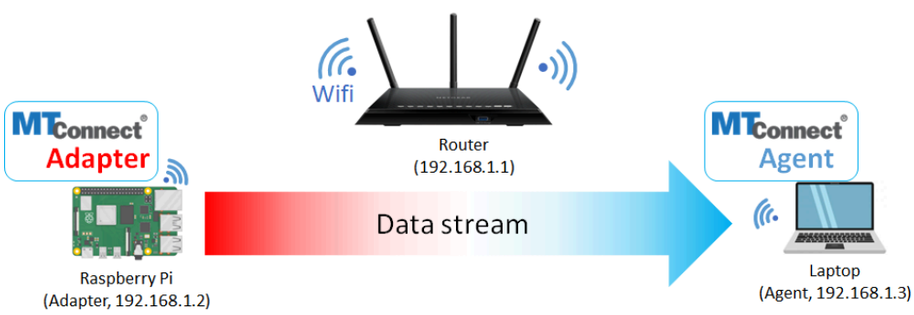


Figure 8 MTConnect schematic for simulation

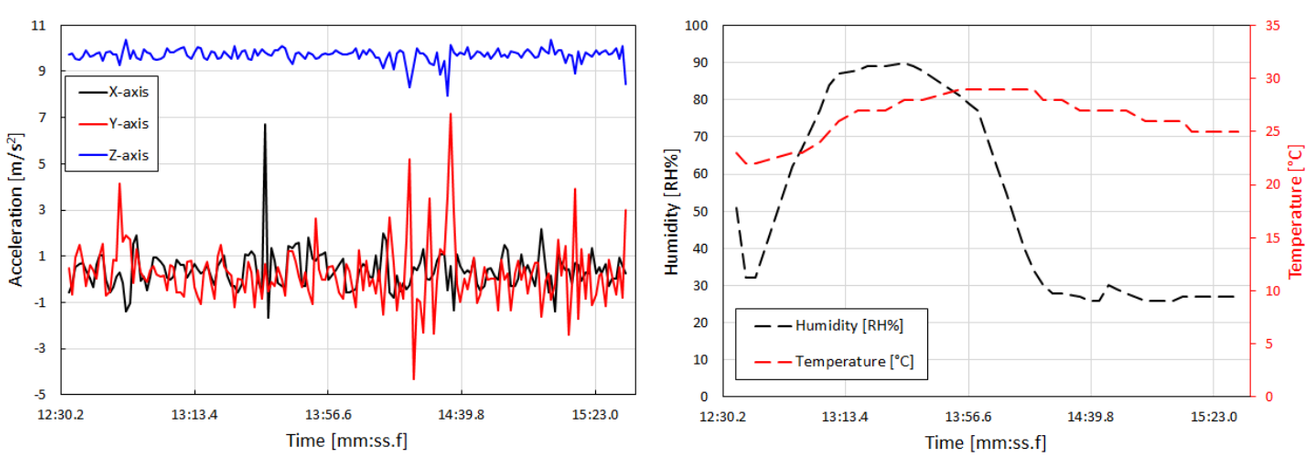


Figure 9 Simulation data ('Sensor_log.txt'): ADXL345 acceleration (left) and DS18B20 & Virtual sensor (right)

First, you need to specify Adapter information in the agent configuration file ('agent.cfg') before running. Please follow the steps below. 1. Open *'agent.cfg'* file of laptop using a text editor. 2. Change appropriate Host and Port in *'Adapters { }'* according to your network environment.

a. Make Port 7878.

b. Host IP address of this example is 192.168.1.2 (your Raspberry Pi IP address must be different). Please make sure of Raspberry Pi's Wifi IP address in the router network.

```
Adapters {
   # Log file has all machines with device name prefixed
   Adapter1 {
      Host = 192.168.1.2
      Port = 7878
   }
}
```

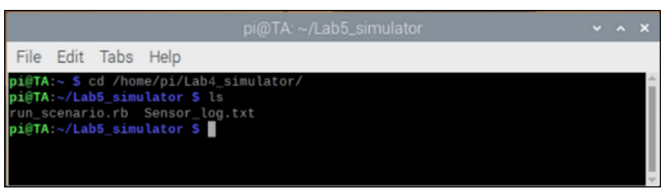
3. Save the file and then close it.
4. Run the agent on laptop.
   a. Please make sure that the agent is running before moving on.

Now, you are ready to run the simulator. Please follow the steps below. 1. Open 'Terminal'. 2. Change directory using *'cd'* <file directory where *'run_scenario.rb' and 'Sensor_log.txt'* are in> command as below.

Raspberry Pi - Terminal

`cd /home/pi/Lab4_simulator/`

- In this example, the file directory is */home/pi/Lab4_simulator/*.
- Your directory might be different.



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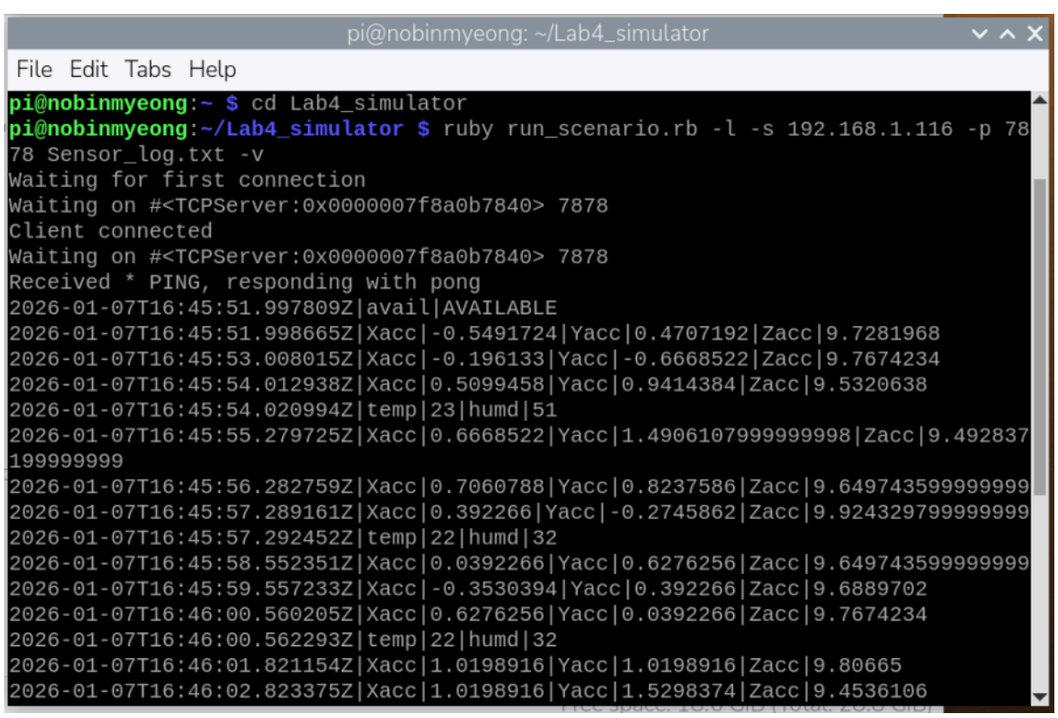
3. Run the simulator with the log file using ruby command as below.

Raspberry Pi - Terminal

`ruby run_scenario.rb -l -s 192.168.1.2 -p 7878 Sensor_log.txt -v`



- The arguments and descriptions are below.
  - ruby : Run Ruby of following *.rb file
  - run_scenario.rb : The MTConnect adapter simulator programed by ruby
  - -l : Loop the file
  - -s : Server IP port to bind to the following IP address
  - 192.168.1.2 : Adapter (here, your PI) IP address
  - -p : Port number of the adapter
  - 7878 : Port number of the adapter

- Sensor_log.txt : Simulation log filename
- -v : Verbose output

- Please make sure of your Adapter IP address.
- Please note that to stop (halt) the simulator, press Ctrl + c on *'Terminal'*.

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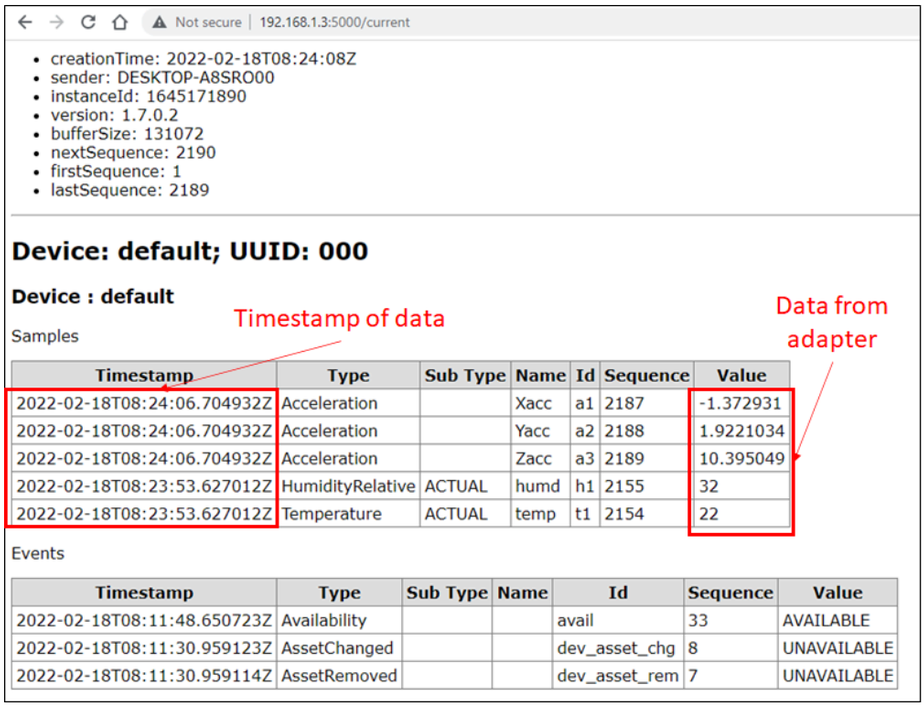
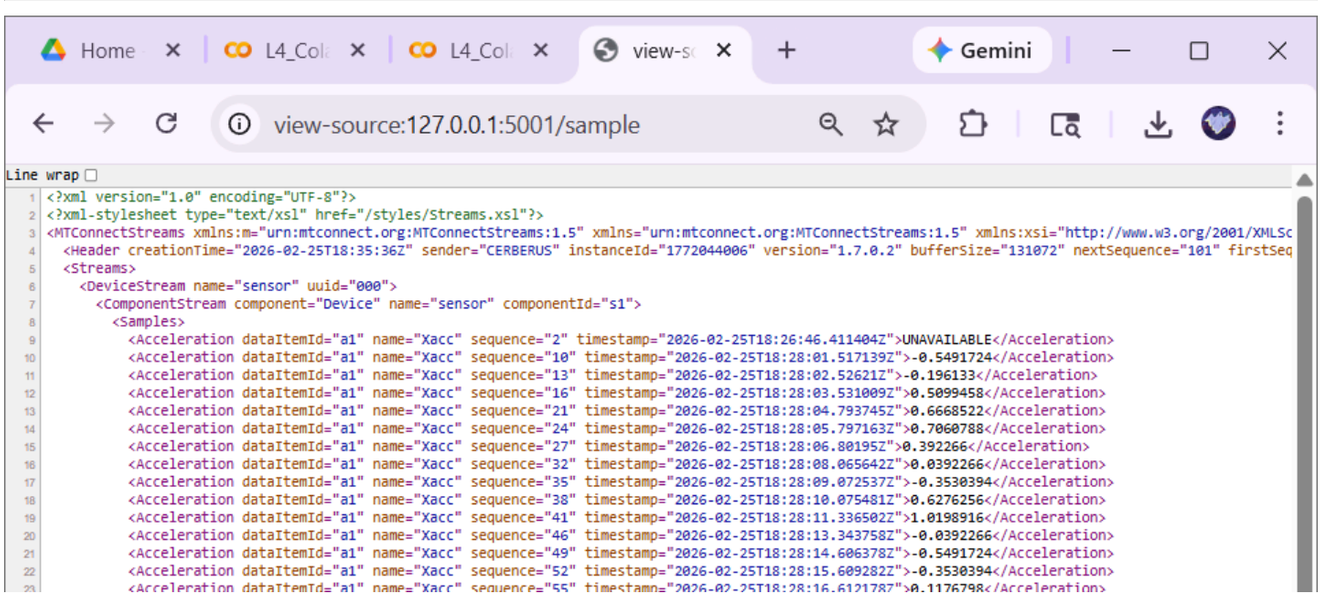
4. Check whether it is working or not by 'current' request data to the agent as below.



Task 3.1

1. Request 'sample' data (laptop: agent, Raspberry Pi: adapter) on a web browser of either Raspberry Pi or laptop and then capture and attach it below.
2. What are your IP addresses (laptop and Raspberry Pi)?



Laptop (MTConnect agent): 192.168.1.13

Raspberry Pi (MTconnect adapter-simulator):192.168.1.23

Task 3.2

1. Repeat Task 3.1 in contrariwise MTConnect configuration as Figure 10, i.e., Raspberry Pi is MTConnect agent and laptop is MTConnect adapter (simulator) in this case.

- The command to run the simulator on *'Command Prompt'* is the same with Raspberry Pi

2. Capture 'sample' data on a web browser of either Raspberry Pi or laptop and then capture and attach it below.
3. Save as 'probe', 'current', 'sample' data from the agent as XML documents (using *'View page source'*) and then attach all as deliverables in Brightspace.

- Filename
    - Lab4_TASK3.2_Probe.xml
    - Lab4_TASK3.2_Current.xml
    - Lab4_TASK3.2_Sample.xml

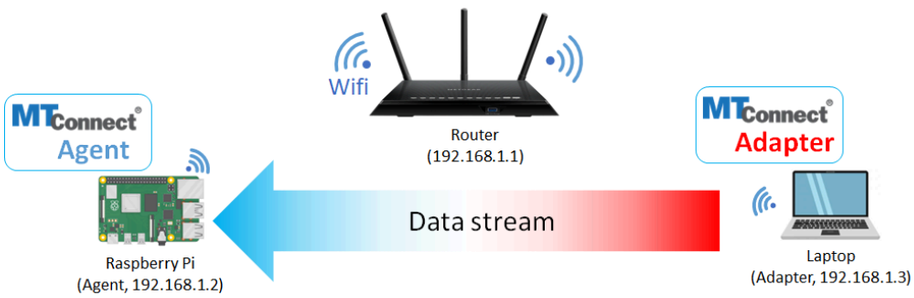
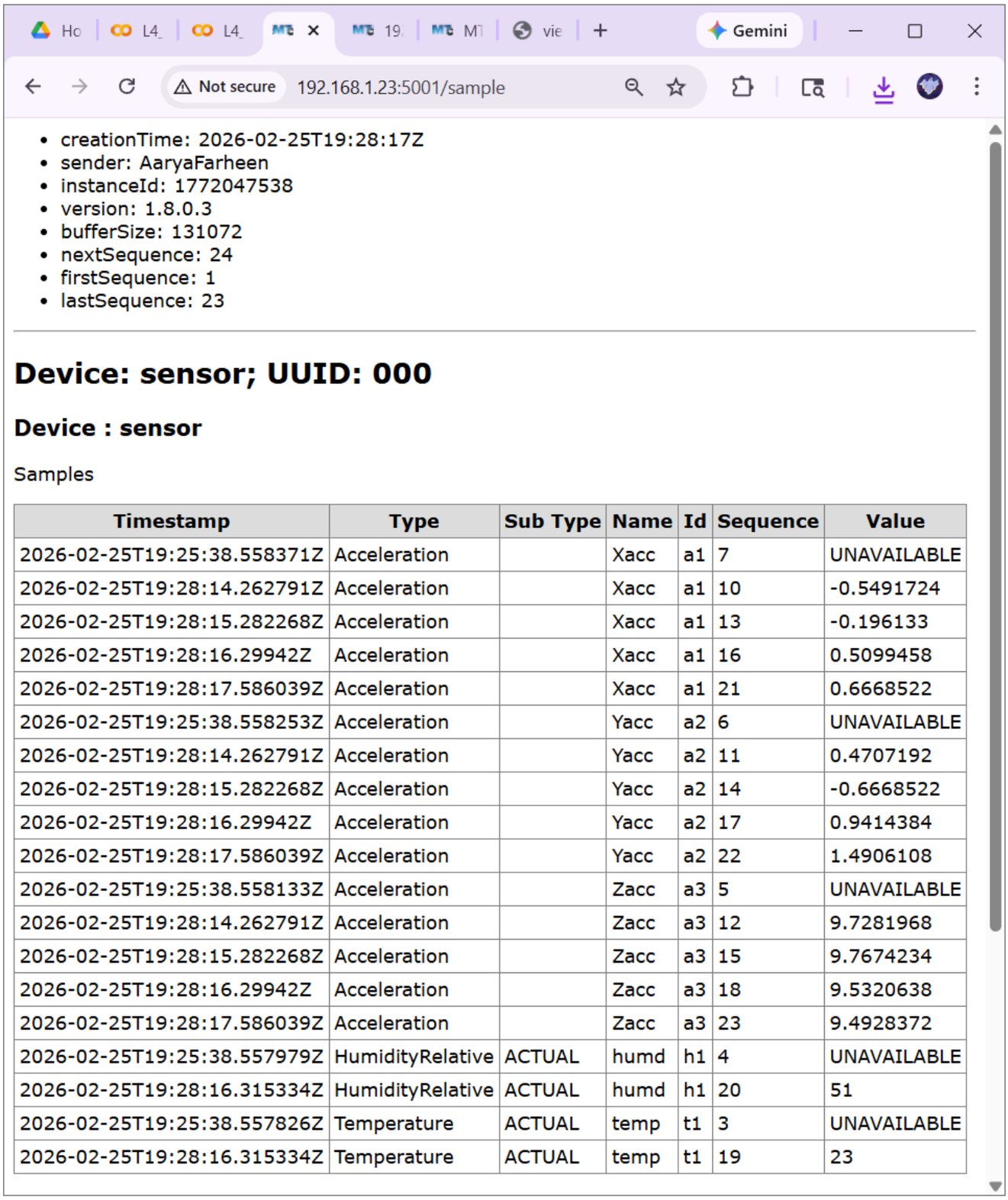


Figure 10 MTConnect schematic for simulation (TASK5)



- creationTime: 2026-02-25T19:28:17Z
- sender: AaryaFarheen
- instanceId: 1772047538
- version: 1.8.0.3
- bufferSize: 131072
- nextSequence: 24
- firstSequence: 1
- lastSequence: 23

Device: sensor; UUID: 000

Device : sensor

Samples

Timestamp	Type	Sub Type	Name	Id	Sequence	Value
2026-02-25T19:25:38.558371Z	Acceleration		Xacc	a1	7	UNAVAILABLE
2026-02-25T19:28:14.262791Z	Acceleration		Xacc	a1	10	-0.5491724
2026-02-25T19:28:15.282268Z	Acceleration		Xacc	a1	13	-0.196133
2026-02-25T19:28:16.29942Z	Acceleration		Xacc	a1	16	0.5099458
2026-02-25T19:28:17.586039Z	Acceleration		Xacc	a1	21	0.6668522
2026-02-25T19:25:38.558253Z	Acceleration		Yacc	a2	6	UNAVAILABLE
2026-02-25T19:28:14.262791Z	Acceleration		Yacc	a2	11	0.4707192
2026-02-25T19:28:15.282268Z	Acceleration		Yacc	a2	14	-0.6668522
2026-02-25T19:28:16.29942Z	Acceleration		Yacc	a2	17	0.9414384
2026-02-25T19:28:17.586039Z	Acceleration		Yacc	a2	22	1.4906108
2026-02-25T19:25:38.558133Z	Acceleration		Zacc	a3	5	UNAVAILABLE
2026-02-25T19:28:14.262791Z	Acceleration		Zacc	a3	12	9.7281968
2026-02-25T19:28:15.282268Z	Acceleration		Zacc	a3	15	9.7674234
2026-02-25T19:28:16.29942Z	Acceleration		Zacc	a3	18	9.5320638
2026-02-25T19:28:17.586039Z	Acceleration		Zacc	a3	23	9.4928372
2026-02-25T19:25:38.557979Z	HumidityRelative	ACTUAL	humd	h1	4	UNAVAILABLE
2026-02-25T19:28:16.315334Z	HumidityRelative	ACTUAL	humd	h1	20	51
2026-02-25T19:25:38.557826Z	Temperature	ACTUAL	temp	t1	3	UNAVAILABLE
2026-02-25T19:28:16.315334Z	Temperature	ACTUAL	temp	t1	19	23

🛠️Troubleshooting - Windows Firewall

When you run ruby simulator on laptop, ‘Windows Security Alert’ may pop up. Allow network communication for ruby as Figure 11.

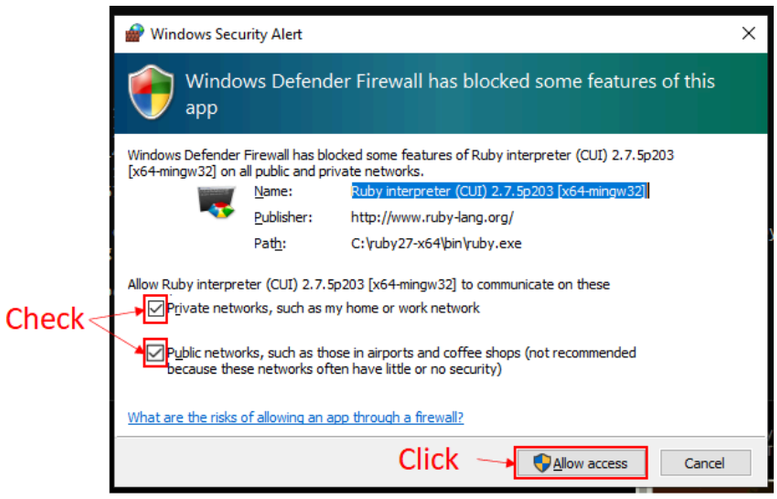
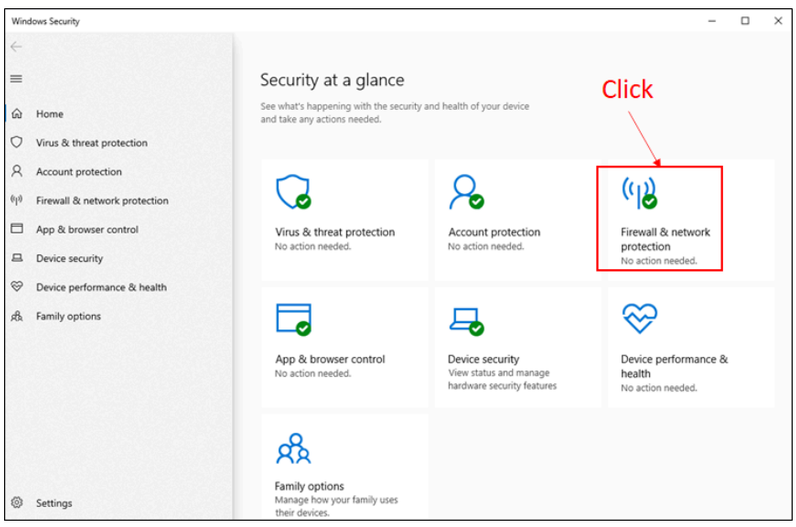


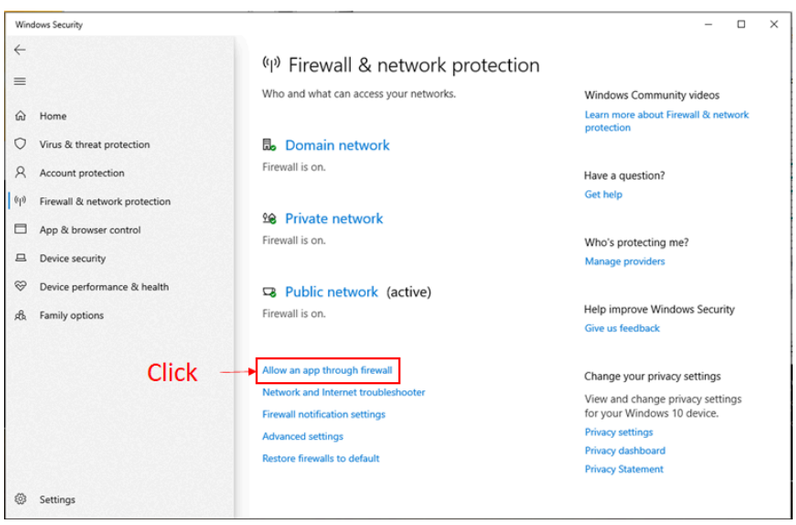
Figure 11 Allow firewall for ruby on Windows

Even if you allow the firewall as above, the simulator (taken a few seconds to run) may not work. If not working, please follow the steps below

1. Open *'Windows Security'* and then click *'Firewall & network protection'*.



2. Click *'Allow on app through firewall'*



3. Click *'Change settings'*.
4. Check all the checkboxes related to *'Ruby Interpreter'* in Allowed apps and features
5. Click *'OK'*.

Task 3.3

1. (**DO THIS AT HOME**) Repeat Task 3.1 by running both the agent and adapter (simulation) on laptop as configuration in Figure 12.
   - Hint: Adapter IP address should be 127.0.0.1.

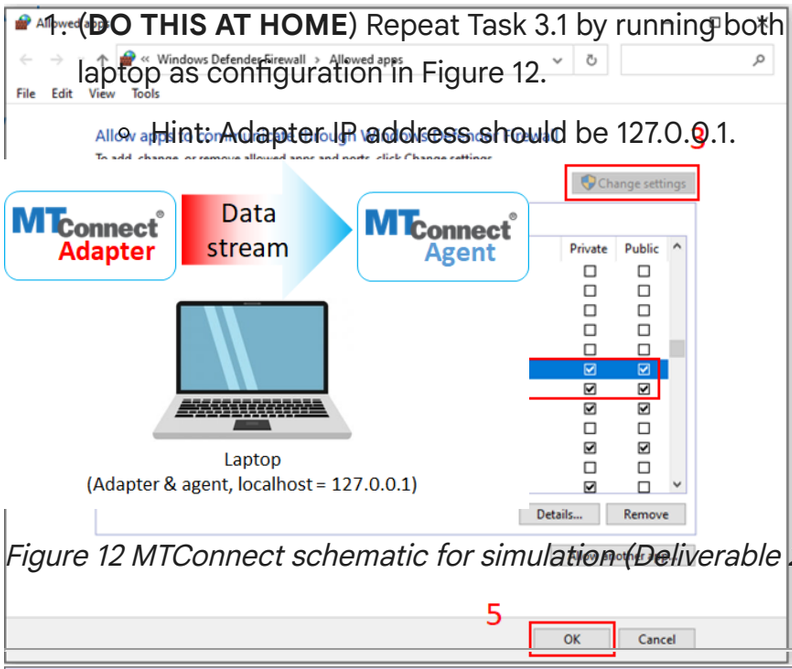
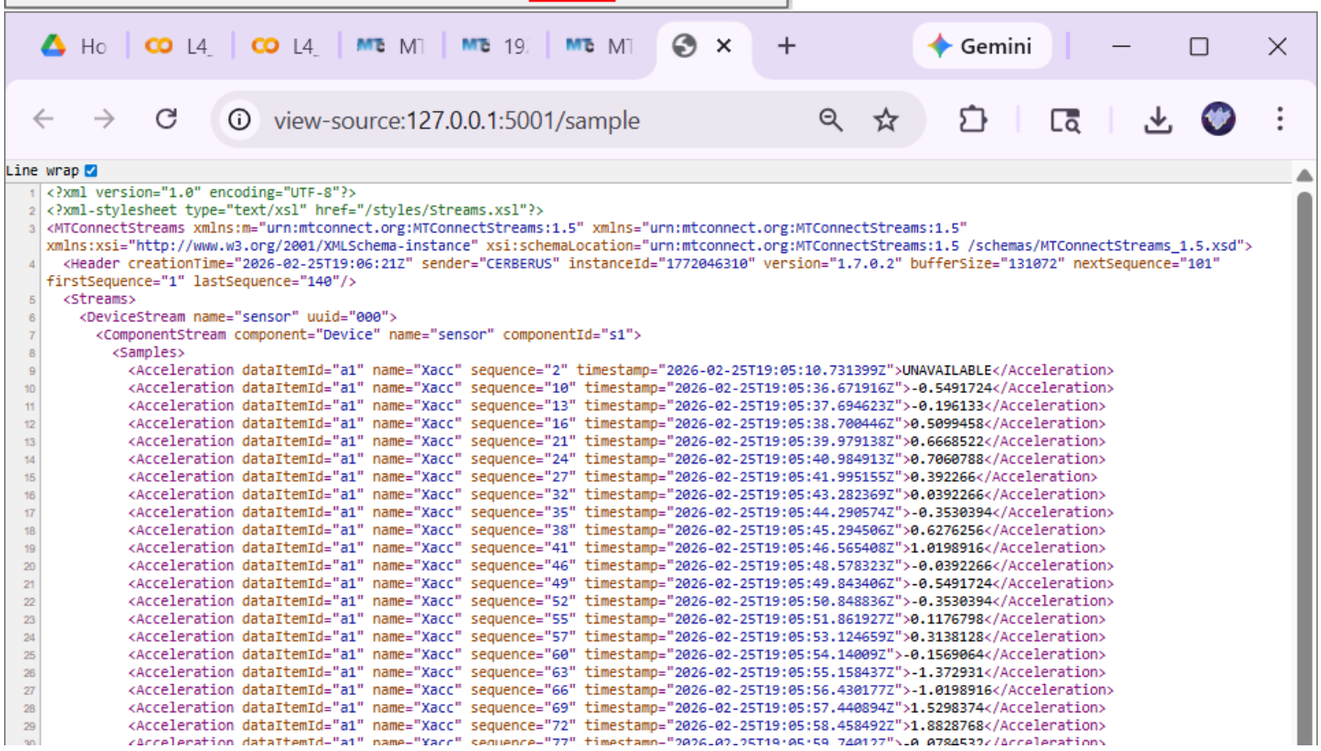


Figure 12 MTConnect schematic for simulation (Deliverable 2)




```xml
<?xml version="1.0" encoding="UTF-8"?>
<?xml-stylesheet type="text/xsl" href="/styles/Streams.xsl"?>
<MTConnectStreams xmlns:m="urn:mtconnect.org:MTConnectStreams:1.5" xmlns="urn:mtconnect.org:MTConnectStreams:1.5"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xsi:schemaLocation="urn:mtconnect.org:MTConnectStreams:1.5 /schemas/MTConnectStreams_1.5.xsd">
  <Header creationTime="2026-02-25T19:06:21Z" sender="CERBERUS" instanceId="1772046310" version="1.7.0.2" bufferSize="131072" nextSequence="101"
firstSequence="1" lastSequence="140"/>
  <Streams>
    <DeviceStream name="sensor" uuid="000">
      <ComponentStream component="Device" name="sensor" componentId="s1">
        <Samples>
          <Acceleration dataItemId="a1" name="Xacc" sequence="2" timestamp="2026-02-25T19:05:10.731399Z">UNAVAILABLE</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="10" timestamp="2026-02-25T19:05:36.671916Z">-0.5491724</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="13" timestamp="2026-02-25T19:05:37.694623Z">-0.196133</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="16" timestamp="2026-02-25T19:05:38.700446Z">0.5099458</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="21" timestamp="2026-02-25T19:05:39.979138Z">0.6668522</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="24" timestamp="2026-02-25T19:05:40.984913Z">0.7060788</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="27" timestamp="2026-02-25T19:05:41.995155Z">0.392266</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="32" timestamp="2026-02-25T19:05:43.282369Z">0.0392266</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="35" timestamp="2026-02-25T19:05:44.290574Z">-0.3530394</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="38" timestamp="2026-02-25T19:05:45.294506Z">0.6276256</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="41" timestamp="2026-02-25T19:05:46.565408Z">1.0198916</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="46" timestamp="2026-02-25T19:05:48.578323Z">-0.0392266</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="49" timestamp="2026-02-25T19:05:49.843406Z">-0.5491724</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="52" timestamp="2026-02-25T19:05:50.848836Z">-0.3530394</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="55" timestamp="2026-02-25T19:05:51.861927Z">0.1176798</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="57" timestamp="2026-02-25T19:05:53.124659Z">0.3138128</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="60" timestamp="2026-02-25T19:05:54.14009Z">-0.1569064</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="63" timestamp="2026-02-25T19:05:55.158437Z">-1.372931</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="66" timestamp="2026-02-25T19:05:56.430177Z">-1.0198916</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="69" timestamp="2026-02-25T19:05:57.440894Z">1.5298374</Acceleration>
          <Acceleration dataItemId="a1" name="Xacc" sequence="72" timestamp="2026-02-25T19:05:58.458492Z">1.8828768</Acceleration>
```

Lab4 Summary and Deliverables

Answer the following questions for your achievements.

Q1. Please summarize Lab4.

Learn to configure MTConnect Agent and Adapter (Simulator) in different network setups.

Capture probe, current, and sample data from the agent in XML format.

Understand agent–adapter communication over network or localhost.

Q2. What skills did you have to develop to accomplish this project?

This project strengthened your networking, XML handling, command-line, simulation, troubleshooting, and data collection skills, while giving you hands-on experience with industrial communication protocols (MTConnect).

Q3. What aspects of this project were the most beneficial for your learning?

MTConnect Agent requires a valid agent.cfg and Device.xml to run.

Network setup matters:

IP 127.0.0.1 → localhost (same device)

Real network IP → remote device access

Adapter simulator sends data to agent via specified IP and port.

Browser /probe, /current, /sample endpoints confirm communication.

Saved XML files provide data logs for analysis and submission.

Q4. What challenges did you encounter in completing the project?

The main challenges revolved around correctly assigning IP addresses, matching ports, ensuring